

Composition Pair B1.07 + B1.16 — The Expression Mechanism as the Governed Downward Pathway for Bidirectional Evolution's Downward Vertical Direction

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This work derives from and formalizes the instinct/reasoning separation pattern introduced in “The Instinct/Reasoning Separation Outside the Model” (Li, April 2026), the second paper in the CKS theory series following “Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems” (Li, April 2026). It does not introduce new axioms. Its contribution is to formalize the enabling relationship between two architectural commitments — B1.07 (expression mechanism) and B1.16 (bidirectional evolution) — as a composition pair: B1.07 provides the governed downward pathway that B1.16’s downward vertical direction requires, and B1.16 provides the evolutionary significance that elevates B1.07 from a one-time deployment configuration into a continuous, architecture-carrying channel.

Abstract

The CKS architecture commits to two mechanisms that, when combined, define how architectural evolution at higher structural levels reaches operational behavior at lower levels. B1.07 (the expression mechanism) governs how DNA-layer substrate elements are activated or deactivated for a given cell goal through a harness substrate that is itself human-governed, inspectable, and modifiable. B1.16 (bidirectional evolution) commits to both horizontal evolution — refinement within existing structure — and vertical evolution across structural levels in two directions: upward, where cells’ operational evidence informs aspect and Self evolution, and downward, where Self and aspect architectural evolution propagates to reshape cell DNA. This note formalizes the enabling pair they form. B1.07 is the governed downward pathway B1.16’s downward vertical direction requires: without it, downward evolution proceeds through direct cell-by-cell DNA modification, carrying governance burden proportional to cell count with no architectural channel. B1.16 is what gives B1.07 its architectural significance as an evolution-carrying channel: without it, expression is used once at deployment initialization and carries no ongoing evolutionary role. Together they complete the downward arm of the vertical evolution architecture, whose upward arm is covered by the symmetric pair B4.08 (B1.15 + B1.16). Both pairs are required for bidirectional vertical evolution to be architecturally complete.

1. Why this pair is the downward arm of vertical evolution

CKS commits to evolution as a continuous architectural property, not a one-time configuration event. Three mechanisms — instinct evolution, DNA evolution, and action-feedback evolution — operate in productive tension at every level of the three-level structure (cell, aspect, Self). When DNA evolution produces a change at the Self level — new integration architecture across aspects, revised instinct/reasoning configuration, updated cross-aspect governance logic — that change is not architecturally useful until it reaches the cells whose operational behavior should reflect it. The question of *how* higher-level architectural evolution propagates downward to operational behavior is not answered by naming evolution as a commitment; it requires an operational pathway.

That pathway is the expression mechanism, B1.07. But the pathway’s significance depends on a prior question: is the pathway used once, at deployment initialization, to establish an initial configuration? Or is it used continuously, as an ongoing channel through which every relevant architectural change at the Self level reaches its cells? The answer to that question is supplied by B1.16’s downward vertical direction: bidirectional evolution commits that architectural evolution at higher levels propagates downward as a continuous architectural event, not a one-time configuration act. The expression mechanism is the channel through which this continuous propagation flows.

These two commitments form an enabling pair. Neither achieves its architectural purpose without the other.

2. B1.07: The expression mechanism

The expression mechanism governs which DNA-layer substrate elements are active for a given cell goal (§5.4 of the source paper). Each cell carries a harness substrate — a governed, human-inspectable, modifiable, and overridable configuration artifact — that selects which sub-substrates are active. Two design patterns are available per deployment: the full-DNA-with-selective-expression pattern, in which the cell carries the complete DNA of its parent Self or aspect with expression governing which elements are activated, and the partial-DNA-slice pattern, in which the cell carries only the subset of DNA elements relevant to its task scope. The choice between these patterns is a per-deployment governance decision under orchestration substrate authority (B2.31, B2.32).

The expression mechanism’s governance properties are direct inheritors of Paper 1’s commitments. The harness substrate that governs expression is substrate content: it is human-governed (inspect, modify, override rights held at all times), tool-agnostic (realizable in any environment supporting persistent structured state with human read/write access), and path-retraceable (changes to expression configuration carry provenance metadata). Expression is not LLM-driven judgment about which DNA elements to activate; the selection logic is governed substrate content, authored and maintained under human authority.

The mechanism’s scope spans all three architectural levels. At the cell level, expression governs which DNA elements a cell activates for a specific task. At the aspect level, the same mechanism governs how

a purpose-defined coordination arrangement over cells selects which DNA elements activate for its operational mode. At the Self level, expression governs how the integrated whole — holding multiple aspects — propagates its DNA configuration to the aspects and cells it comprises. The mechanism is recursive in the same way Paper 2's architectural commitments are recursive: the same machinery operates at each level without requiring a new primitive per level (B2.33, B2.34).

3. B1.16: Bidirectional evolution's vertical dimension

Bidirectional evolution as formalized in B1.16 names two axes and two directions within the vertical axis (§7.4 of the source paper). Horizontal evolution refines content within existing structure: cells refine their DNA, aspects refine their cell composition, Selves refine their aspect arrangements. Vertical evolution reorganizes structure across levels: cells split or merge, aspects gain or lose constituent cells, Selves gain or lose aspects, relational roles reconfigure.

The vertical axis is bidirectional. The upward direction carries evidence from operational behavior at lower levels upward to inform architectural decisions at higher levels — cells' accumulated action evidence feeds into aspect and Self DNA evolution through action-feedback mechanisms (B1.15 + B1.16, covered in B4.08). The downward direction carries architectural decisions at higher levels downward to reshape operational behavior at lower levels — Self and aspect DNA evolution propagates to cells through the expression mechanism (the subject of this note).

Two properties distinguish CKS vertical evolution from biological analogues and from prior software-architecture evolution frameworks. First, the substrates that define composition — the orchestration rules and relational roles that specify what a Self is, how aspects are arranged, which cells belong to which aspects — are themselves substrate content, governed under the same authority architecture as DNA content. Structural reorganization is therefore a substrate-edit operation, not a development-program rewrite or graph-genome manipulation. Second, both horizontal and vertical evolution operate through the three mechanisms (instinct evolution, DNA evolution, action-feedback evolution) at every level, and the mechanisms are architecturally separated rather than collapsed into a single operator. The downward vertical direction is not a separate fourth mechanism; it is the directional consequence of DNA evolution at higher levels, and its operational realization is the expression pathway B1.07 provides (B2.81).

4. The enabling direction: B1.07 enables B1.16's downward vertical

When Self DNA evolves through directed selection — new integration architecture, revised instinct/reasoning configuration per B2.23, updated cross-aspect governance rules — those changes must reach cells for the evolution to be operationally realized. The expression mechanism is the governed pathway for this propagation (B2.31).

The enabling relationship is precise. Self governance per B1.14 produces Self DNA changes. The Self DNA includes updated harness substrate configuration per B2.30. The expression mechanism activates

or deactivates cell DNA elements per B2.31 to reflect Self DNA changes. The carry-strategy per B2.32 determines how extensively Self DNA changes propagate — whether cells carry the full updated DNA with new expression settings, or whether they carry updated partial slices. Downward vertical evolution per B2.81 is thereby realized: cells' operational behavior reflects Self-level architectural evolution without requiring direct modification of each cell's DNA individually.

The alternative pathway — absent B1.07 — is direct cell-by-cell DNA modification. Each cell whose operational behavior should reflect a Self-level architectural change must be individually identified, its DNA directly accessed, and its elements individually modified under whatever orchestration rules govern direct cell modification. The governance burden of this pathway scales with cell count: a Self with many cells undergoes a modification cost proportional to the number of affected cells for every Self-level architectural change. No governed architectural channel exists through which Self-level reasoning about the change propagates to cells as a coherent unit; each cell modification is an independent governance event. The expression mechanism eliminates this burden by creating the governed channel: the Self-level change is expressed once through the harness substrate configuration, and the cells whose DNA is governed by that harness substrate reflect the change through their governed selection machinery.

Without B1.07, B1.16's downward direction is operationally present as an architectural aspiration but lacks the governed pathway to realize it. The commitment that Self DNA evolution propagates to cell operational behavior has no channel through which it flows except direct cell-by-cell modification — a pattern that does not scale and does not preserve the governance coherence that architecture-level evolution requires.

5. The motivating direction: B1.16 gives B1.07 its evolution-carrying significance

The expression mechanism, considered without B1.16's continuous downward evolution, is a deployment configuration feature. It is invoked at initialization: when a Self is deployed, the harness substrate configuration determines which DNA elements activate for which cells and which operational modes. After initialization, the expression configuration is stable until it is manually changed. The mechanism has done its work at deployment time; it does not play a recurring architectural role in the deployment's ongoing operation.

B1.16's downward vertical direction transforms this. When Self DNA undergoes architectural evolution — not as a one-time initialization event but as an ongoing architectural event driven by DNA evolution per B1.14 — every change that should propagate to cell behavior flows through the expression pathway. New integration architecture between aspects requires updated expression configuration at the aspect-cell boundary. Revised instinct/reasoning configuration per B2.23 — repositioning the boundary between what cells handle through instinct and what they handle through governed substrate reasoning — requires updated expression to reflect the repositioned boundary in cell DNA activation. Updated cross-aspect governance rules, when they affect which DNA elements are operative for cells that participate in multiple aspects, require updated expression to carry those

governance changes downward.

The expression mechanism, with B1.16, is invoked not once but continuously — every time Self-level architectural evolution produces a change that should propagate to cell operational behavior. Expression becomes a live architectural channel: the medium through which the deployment's Self-level architectural intelligence reaches its operational units. This is B1.16's contribution to B1.07: not a technical modification to how expression works, but the architectural context that makes expression's ongoing use architecturally significant rather than an incidental reconfiguration event. B1.16 does not make B1.07 technically possible; B1.07 already exists as a governed selection mechanism. B1.16 makes B1.07 architecturally indispensable: a deployment that evolves continuously through B1.16's downward direction must use the expression mechanism continuously as the channel for that evolution.

6. Mutual dependency, co-presence requirement, and failure modes

Mutual dependency. B1.07 depends on B1.16 for operational frequency: without B1.16 as the motivating direction, expression is invoked once and is thereafter architecturally inert. B1.16's continuous downward evolution transforms expression from a one-time event into a live channel with ongoing architectural significance. B1.16 depends on B1.07 for downward mechanism: the downward vertical direction of bidirectional evolution requires an operational pathway to realize the propagation it commits to. B1.07 is that pathway. Without B1.07, the downward direction of B1.16 proceeds through direct cell-by-cell modification — a governance-burden scaling pattern that defeats the architectural purpose of having a governed propagation channel.

Co-presence states. Three states are architecturally distinct. First: B1.07 without B1.16. The expression mechanism exists and governs cell DNA activation, but there is no ongoing downward evolution for it to carry. Expression serves deployment initialization; Self DNA evolves through B1.14 but the expression mechanism is not the channel through which that evolution propagates to cells. Second: B1.16 without B1.07. Bidirectional evolution is committed to at the architectural level but the governed pathway for downward propagation is absent. Self-level DNA changes reach cells only through direct cell-by-cell modification, carrying governance burden proportional to cell count for every Self-level change. Third: full co-presence. Self DNA evolution flows downward through the expression mechanism to cell DNA in a governed architectural channel. Downward vertical evolution is efficient, governed, and traceable.

Detection. B2.34 (expression mechanism verification) asks: is the downward pathway functional — do changes to Self DNA produce corresponding changes in cell DNA activation through the expression channel rather than through direct cell DNA modification? B2.83 (bidirectional evolution verification) asks: is the downward vertical evolution pathway configured? Pair completeness asks: do Self DNA changes reach cell operational behavior through expression, or do they require direct cell-by-cell DNA modification?

Failure modes. Expression Mechanism Bypass (anti-pattern B3.08) occurs when B1.07 is absent or inoperative: Self-level architectural evolution proceeds through direct cell DNA modification, governance burden scales with cell count, and no governed downward channel exists. Unidirectional Evolution (anti-pattern B3.17) occurs when B1.16's downward direction is absent: the deployment can receive evidence upward through action-feedback but cannot push architectural changes downward through a governed channel. Both failure modes produce a deployment that is architecturally incomplete with respect to vertical evolution.

7. Symmetric relationship with B4.08 and completeness of bidirectional vertical evolution

This note is the downward complement of B4.08, which formalizes the B1.15 + B1.16 enabling pair for the upward vertical direction. The two pairs together define the complete bidirectional vertical evolution architecture. B4.08 establishes: action-feedback carries operational evidence from cells upward to inform aspect and Self DNA evolution. Cells that have accumulated evidence through their action layers feed that evidence through the action-feedback mechanism (B1.15) into governed proposals for aspect and Self DNA refinement. The upward direction closes the feedback loop from operational experience to architectural decision. B4.09 establishes: expression carries aspect and Self architectural evolution downward to reshape cell DNA. When aspect and Self DNA evolves through directed selection — incorporating the feedback the upward direction brought — the expression mechanism (B1.07) carries those architectural decisions down to the cells whose operational behavior should reflect them.

The two directions are functionally interdependent. Cells produce evidence that informs upward evolution (through B1.15). Upward evolution produces Self-level architectural changes. Those changes propagate downward to cells (through B1.07). Cells then operate under the updated architecture, producing new evidence, closing the loop. A deployment with B1.16 but only B1.15 (no B1.07) can receive evidence upward but cannot efficiently push architectural changes downward. A deployment with B1.16 but only B1.07 (no B1.15) can push changes downward but cannot receive evidence upward. Both pairs together are required for bidirectional vertical evolution to be architecturally complete.

This pair also stands in contrast to B4.03, which formalizes B1.01 + B1.07 (instinct/reasoning separation and expression mechanism). B4.03 covers expression as the cross-level pathway through which the instinct/reasoning separation manifests at cell operational behavior: how the governed boundary between what cells handle through instinct and what they handle through substrate reasoning is expressed into cell DNA activation. B4.09 covers expression as the downward evolution carrier specifically: how ongoing Self-level architectural evolution is continuously propagated to cells through the expression channel. The expression mechanism is invoked by both pairs, but for architecturally distinct purposes: B4.03 is about the structural boundary; B4.09 is about the evolutionary pathway.

8. Operational test

A deployment instantiates the B1.07 + B1.16 enabling pair if and only if all of the following are true:

1. **Expression is the downward channel.** When Self DNA evolves through directed selection — new integration architecture, revised instinct/reasoning configuration, updated cross-aspect governance — cell DNA elements are updated through the expression mechanism (harness substrate activation/deactivation per B2.31) rather than through direct cell-by-cell DNA modification.
2. **Expression is invoked continuously.** The expression mechanism is not limited to deployment initialization. When Self-level architectural evolution produces changes that should propagate to cells, those changes flow through the expression pathway as a continuous architectural event.
3. **Downward propagation is governed.** The harness substrate that governs expression is substrate content: human-inspectable, modifiable, and overridable at all times. The carry-strategy per B2.32 is itself an orchestration-substrate decision under human governance authority.
4. **Governance burden does not scale with cell count.** A Self-level architectural change that should propagate to all affected cells does so through the expression channel — not through individual cell modification events that scale with cell count. The governance cost of downward propagation is proportional to the complexity of the change, not the number of cells it affects.
5. **Bidirectional vertical evolution is architecturally complete.** The deployment instantiates both B4.08 (action-feedback as upward pathway, B1.15 + B1.16) and B4.09 (expression as downward pathway, B1.07 + B1.16). Detection per B2.34 (expression mechanism verification) and B2.83 (bidirectional evolution verification) are both operative.

A deployment that routes Self-level architectural changes through direct cell DNA modification rather than through the expression channel fails condition 1 (Expression Mechanism Bypass). A deployment in which expression is present but used only at initialization, with subsequent Self-level evolution bypassing expression, fails condition 2. A deployment in which only the upward direction (B4.08) is present without the downward direction (B4.09) fails condition 5.

9. Source paper citation

Li, Wenxin. “The Instinct/Reasoning Separation Outside the Model: Extending the Coordination Knowledge Substrate Pattern to AI Selves under Human Governance.” Independent publication, April 2026. Relevant sections: §5.4 (expression mechanism), §7.4 (dual-axis evolution — horizontal and vertical), §7.3 (multi-level simultaneous evolution), §8 (multi-shaped human governance across evolution mechanisms).

Li, Wenxin. “Coordination Outside the Model: A Human-Governed Substrate Pattern for AI Systems.” Independent publication, April 2026. Relevant sections: §3.3 (authority-not-labor), §4.1 (substrate/LLM governance boundary), §6 (linear-cost scaling).

10. Series self-citation

This note is part of the CKS Derivation Notes series, Phase B4 (composition pairs). Related notes in this series:

- **B4.08** — Composition Pair B1.15 + B1.16: action-feedback as the governed upward pathway for bidirectional evolution's upward vertical direction (symmetric pair; together with B4.09 completes bidirectional vertical evolution).
- **B4.03** — Composition Pair B1.01 + B1.07: expression mechanism as the instinct/reasoning cross-level pathway (related pair; covers expression's role in the structural boundary rather than in ongoing evolutionary propagation).
- **B2.30–B2.34** — Expression mechanism decomposition: the harness substrate as expression governor; full-DNA-with-selective-expression pattern; partial-DNA-slice pattern; per-deployment design choice under orchestration substrate; where expression exceeds biology.
- **B2.79–B2.83** — Bidirectional evolution decomposition: mutation governance; directed selection governance; action-feedback governance; the instinct/reasoning boundary as governance substrate content; high-stakes decision pinning to reasoning layer.
- **B2.81** — Vertical evolution (both directions): the commitment that evolution operates across structural levels in both upward and downward directions.